DocChat AI Architecture Overview - Page 1

DocChat AI is an enterprise-grade document intelligence platform powered by Retrieval-Augmented Generation (RAG). In

The DocumentLoader service is responsible for ingesting files from in-memory byte buffers or the local filesystem. For Port

DocChat AI Chunking & Embedding Pipeline - Page 2

Text chunking is executed by the TextChunker service. Rather than arbitrarily slicing strings at rigid byte boundaries, the cl

This overlap prevents semantic cliff effects where critical context or named entities are split across retrieval boundaries. Fu

Each generated chunk is enriched with essential metadata: doc_id, filename, source page number, global chunk index, an

DocChat AI Vector Storage & Persistence - Page 3

Once vector embeddings are generated, chunks and their metadata are committed to a persistent ChromaDB collection via

The ChromaVectorStore exposes a clean public interface featuring add_chunks() for upserting vectorized content, list_doc

This ingestion architecture guarantees high retrieval accuracy, sub-50ms vector query latencies, and deterministic isolation